

Maxence Debes

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MSc Data Science student with a strong academic background in mathematics and computer science, specialized in Machine Learning and AI — combining mathematical rigor with advanced Python to build robust, data-driven solutions.

Projects

Hi! PARIS Hi!ckathon 2025 (2nd place)

- Ranked 2nd out of 80+ teams in the 2025 edition of the Hi!Paris data science hackathon. Task: Predicting student PISA scores from complex socio-economic backgrounds.
- Engineered a Gated Model (router composed of two XGBoost regressors), achieving an R^2 score of 0.79 on the final test set.

Molecular Graph Captioning (Kaggle Competition)

- Development of a retrieval-based architecture using contrastive learning to align molecular graph embeddings with natural language descriptions in a joint latent space.
- Implementation of a dedicated graph neural network encoder using PyTorch Geometric, optimized for semantic alignment through BERTScore and BLEU-4 metrics.

Streaming Safety Classifier for LLMs — Capstone Project with DragonLLM

- Reproduced the toxicity classifier from the Qwen3Guard technical report: synthesized diverse toxic and safe conversations with Qwen3-30B and labeled them using Qwen3Guard's safety classifier.
- Trained a lightweight per-token safety head on top of a frozen Qwen backbone for real-time, streaming unsafe-content detection.

Data Privacy & Anonymization — Project Cassiopee

- Assessed privacy and re-identification risks in anonymized datasets through OSINT case studies, and proposed data-protection strategies. Conducted at Télécom SudParis.

Automated Parking Optimization

- Developed a Python optimization model for vehicle placement in multi-level automated parking, combining A* search and simulated annealing to minimize retrieval costs and improve efficiency.

Work Experience

Data Science Intern — ChemAI

Apr 2026 — present

- Building and analyzing a graph database of chemical reactions to extract actionable insights.

IT Data Governance Intern — Batigere Maison Familiale

Jul 2025 — Aug 2025

- Optimized digital infrastructure by migrating the company to SharePoint in order to enhance collaboration and document management, with adoption by 100% of employees.
- Designed and implemented a GDPR compliance policy, reinforcing data security and ensuring regulatory compliance.

Software Engineer Intern — Atos

Jul 2024 — Aug 2024

- Designed and implemented a telework tracking application using the JHipster application generator to facilitate accessibility to workstations.

Education

École Polytechnique — Polytechnic Institute of Paris

2025 — present

MSc Data Science

Relevant coursework: ALTEGRAD (Joint MVA Course), Deep Learning, Reinforcement Learning, Representation Learning (MVA), Generative Models, Optimization for Data Science.

Télécom SudParis — Polytechnic Institute of Paris

2023 — present

Master's Degree in Computer Science and Applied Mathematics

Relevant coursework: Statistics and Data Analysis, Optimization, Machine Learning and Data Mining, Stochastic Processes, Continuous Optimization for Learning.

Skills & Interests

Technical Skills: Python, PyTorch, Transformers, Scikit-learn, Pandas, SQL, Java, Git, Linux, $\LaTeX$

Languages: French (native), English (C1 — TOEIC 965/990), German (A2)

Engagement & Interests: Former President of student association; Handball (Competitive) & Music